

Q. 1 – Q. 5 carry one mark each.

Q.1 The man who is now Municipal Commissioner worked as _____.

- (A) the security guard at a university
- (B) a security guard at the university
- (C) a security guard at university
- (D) the security guard at the university

Q.2 Nobody knows how the Indian cricket team is going to cope with the difficult and seamer-friendly wickets in Australia.

Choose the option which is closest in meaning to the underlined phrase in the above sentence.

- (A) put up with (B) put in with (C) put down to (D) put up against

Q.3 Find the odd one in the following group of words.

mock, deride, praise, jeer

- (A) mock (B) deride (C) praise (D) jeer

Q.4 Pick the odd one from the following options.

- (A) CADBE (B) JHKIL (C) XZYWZ (D) ONPMQ

Q.5 In a quadratic function, the value of the product of the roots (α, β) is 4. Find the value of

$$\frac{\alpha^n + \beta^n}{\alpha^{-n} + \beta^{-n}}$$

- (A) n^4 (B) 4^n (C) 2^{2n-1} (D) 4^{n-1}

Q. 6 – Q. 10 carry two marks each.

Q.6 Among 150 faculty members in an institute, 55 are connected with each other through Facebook® and 85 are connected through WhatsApp®. 30 faculty members do not have Facebook® or WhatsApp® accounts. The number of faculty members connected only through Facebook® accounts is _____.

- (A) 35 (B) 45 (C) 65 (D) 90

- Q.7 Computers were invented for performing only high-end useful computations. However, it is no understatement that they have taken over our world today. The internet, for example, is ubiquitous. Many believe that the internet itself is an unintended consequence of the original invention. With the advent of mobile computing on our phones, a whole new dimension is now enabled. One is left wondering if all these developments are good or, more importantly, required.

Which of the statement(s) below is/are logically valid and can be inferred from the above paragraph?

- (i) The author believes that computers are not good for us.
- (ii) Mobile computers and the internet are both intended inventions

(A) (i) only (B) (ii) only (C) both (i) and (ii) (D) neither (i) nor (ii)

- Q.8 All hill-stations have a lake. Ooty has two lakes.

Which of the statement(s) below is/are logically valid and can be inferred from the above sentences?

- (i) Ooty is not a hill-station.
- (ii) No hill-station can have more than one lake.

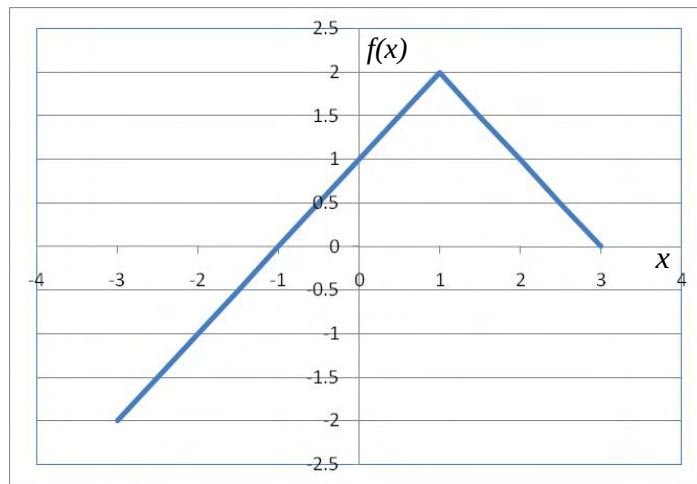
(A) (i) only (B) (ii) only
(C) both (i) and (ii) (D) neither (i) nor (ii)

- Q.9 In a 2×4 rectangle grid shown below, each cell is a rectangle. How many rectangles can be observed in the grid?



(A) 21 (B) 27 (C) 30 (D) 36

Q.10



Choose the correct expression for $f(x)$ given in the graph.

(A) $f(x) = 1 - |x - 1|$

(B) $f(x) = 1 + |x - 1|$

(C) $f(x) = 2 - |x - 1|$

(D) $f(x) = 2 + |x - 1|$

END OF THE QUESTION PAPER

Q. 1 – Q. 25 carry one mark each.

Q.1 The maximum value attained by the function $f(x) = x(x-1)(x-2)$ in the interval $[1, 2]$ is ____.

Q.2 Consider a 3×3 matrix with every element being equal to 1. Its only non-zero eigenvalue is ____.

Q.3 The Laplace Transform of $f(t) = e^{2t} \sin(5t) u(t)$ is

- (A) $\frac{5}{s^2-4s+29}$ (B) $\frac{5}{s^2+5}$ (C) $\frac{s-2}{s^2-4s+29}$ (D) $\frac{5}{s+5}$

Q.4 A function $y(t)$, such that $y(0) = 1$ and $y(1) = 3e^{-1}$, is a solution of the differential equation $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + y = 0$. Then $y(2)$ is

- (A) $5e^{-1}$ (B) $5e^{-2}$ (C) $7e^{-1}$ (D) $7e^{-2}$

Q.5 The value of the integral

$$\oint_C \frac{2z+5}{\left(z-\frac{1}{2}\right)(z^2-4z+5)} dz$$

over the contour $|z| = 1$, taken in the anti-clockwise direction, would be

- (A) $\frac{24\pi i}{13}$ (B) $\frac{48\pi i}{13}$ (C) $\frac{24}{13}$ (D) $\frac{12}{13}$

Q.6

The transfer function of a system is $\frac{Y(s)}{R(s)} = \frac{s}{s+2}$. The steady state output $y(t)$ is $A \cos(2t + \phi)$ for the input $\cos(2t)$. The values of A and ϕ , respectively are

- (A) $\frac{1}{\sqrt{2}}, -45^\circ$ (B) $\frac{1}{\sqrt{2}}, +45^\circ$ (C) $\sqrt{2}, -45^\circ$ (D) $\sqrt{2}, +45^\circ$

Q.7

The phase cross-over frequency of the transfer function $G(s) = \frac{100}{(s+1)^3}$ in rad/s is

- (A) $\sqrt{3}$ (B) $\frac{1}{\sqrt{3}}$ (C) 3 (D) $3\sqrt{3}$

- Q.8 Consider a continuous-time system with input $x(t)$ and output $y(t)$ given by

$$y(t) = x(t) \cos(t)$$

This system is

- (A) linear and time-invariant
- (B) non-linear and time-invariant
- (C) linear and time-varying
- (D) non-linear and time-varying

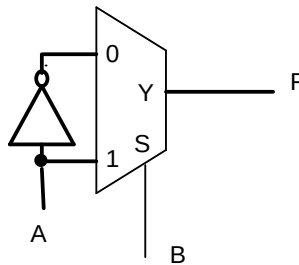
- Q.9 The value of $\int_{-\infty}^{+\infty} e^{-t} \delta(2t - 2) dt$, where $\delta(t)$ is the Dirac delta function, is

- (A) $\frac{1}{2e}$
- (B) $\frac{2}{e}$
- (C) $\frac{1}{e^2}$
- (D) $\frac{1}{2e^2}$

- Q.10 A temperature in the range of -40°C to 55°C is to be measured with a resolution of 0.1°C . The minimum number of ADC bits required to get a matching dynamic range of the temperature sensor is

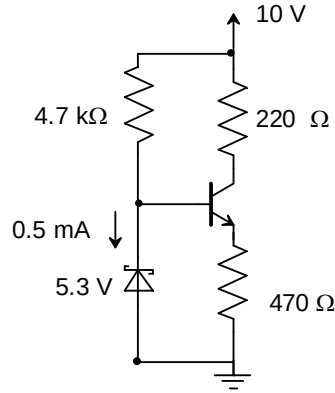
- (A) 8
- (B) 10
- (C) 12
- (D) 14

- Q.11 Consider the following circuit which uses a 2-to-1 multiplexer as shown in the figure below. The Boolean expression for output F in terms of A and B is



- (A) $A \oplus B$
- (B) $\overline{A + B}$
- (C) $A + B$
- (D) $\overline{A \oplus B}$

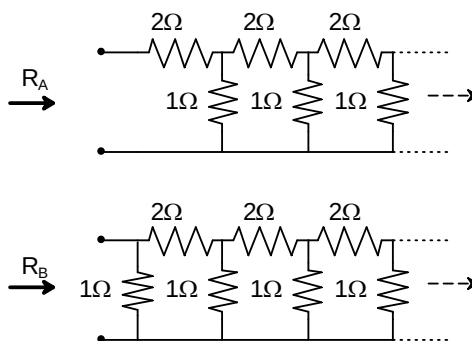
- Q.12 A transistor circuit is given below. The Zener diode breakdown voltage is 5.3 V as shown. Take base to emitter voltage drop to be 0.6 V. The value of the current gain β is _____.



- Q.13 In cylindrical coordinate system, the potential produced by a uniform ring charge is given by $\varphi = f(r, z)$, where f is a continuous function of r and z . Let $\vec{E}$ be the resulting electric field. Then the magnitude of $\nabla \times \vec{E}$
- (A) increases with r . (B) is 0. (C) is 3. (D) decreases with z .

- Q.14 A soft-iron toroid is concentric with a long straight conductor carrying a direct current I . If the relative permeability μ_r of soft-iron is 100, the ratio of the magnetic flux densities at two adjacent points located just inside and just outside the toroid, is _____.

- Q.15 R_A and R_B are the input resistances of circuits as shown below. The circuits extend infinitely in the direction shown. Which one of the following statements is TRUE?

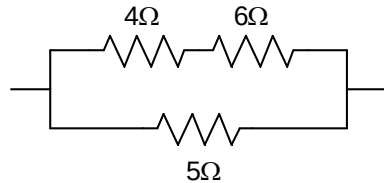


- (A) $R_A = R_B$ (B) $R_A = R_B = 0$ (C) $R_A < R_B$ (D) $R_B = R_A / (1 + R_A)$

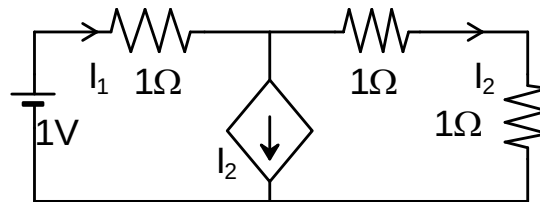
Q.16 In a constant V/f induction motor drive, the slip at the maximum torque

- (A) is directly proportional to the synchronous speed.
- (B) remains constant with respect to the synchronous speed.
- (C) has an inverse relation with the synchronous speed.
- (D) has no relation with the synchronous speed.

Q.17 In the portion of a circuit shown, if the heat generated in $5\ \Omega$ resistance is 10 calories per second, then heat generated by the $4\ \Omega$ resistance, in calories per second, is _____.



Q.18 In the given circuit, the current supplied by the battery, in ampere, is _____.



Q.19 In a 100 bus power system, there are 10 generators. In a particular iteration of Newton Raphson load flow technique (in polar coordinates), two of the PV buses are converted to PQ type. In this iteration,

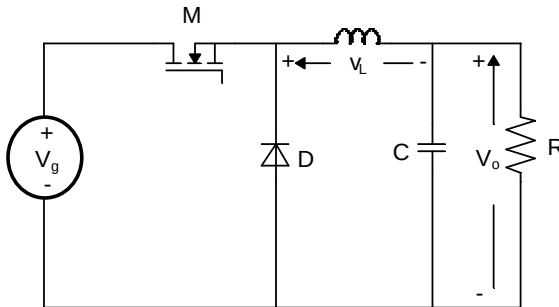
- (A) the number of unknown voltage angles increases by two and the number of unknown voltage magnitudes increases by two.
- (B) the number of unknown voltage angles remains unchanged and the number of unknown voltage magnitudes increases by two.
- (C) the number of unknown voltage angles increases by two and the number of unknown voltage magnitudes decreases by two.
- (D) the number of unknown voltage angles remains unchanged and the number of unknown voltage magnitudes decreases by two.

Q.20 The magnitude of three-phase fault currents at buses A and B of a power system are 10 pu and 8 pu, respectively. Neglect all resistances in the system and consider the pre-fault system to be unloaded. The pre-fault voltage at all buses in the system is 1.0 pu. The voltage magnitude at bus B during a three-phase fault at bus A is 0.8 pu. The voltage magnitude at bus A during a three-phase fault at bus B, in pu, is _____.

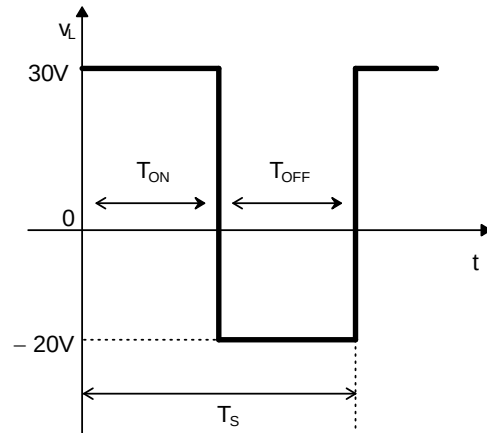
Q.21 Consider a system consisting of a synchronous generator working at a lagging power factor, a synchronous motor working at an overexcited condition and a directly grid-connected induction generator. Consider capacitive VAR to be a source and inductive VAR to be a sink of reactive power. Which one of the following statements is TRUE?

- (A) Synchronous motor and synchronous generator are sources and induction generator is a sink of reactive power.
- (B) Synchronous motor and induction generator are sources and synchronous generator is a sink of reactive power.
- (C) Synchronous motor is a source and induction generator and synchronous generator are sinks of reactive power.
- (D) All are sources of reactive power.

Q.22 A buck converter, as shown in Figure (a) below, is working in steady state. The output voltage and the inductor current can be assumed to be ripple free. Figure (b) shows the inductor voltage v_L during a complete switching interval. Assuming all devices are ideal, the duty cycle of the buck converter is _____.

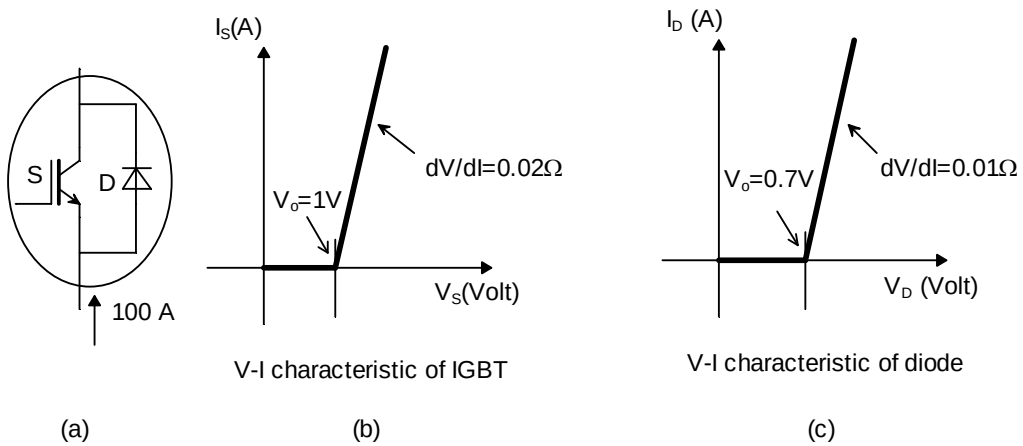


(a)

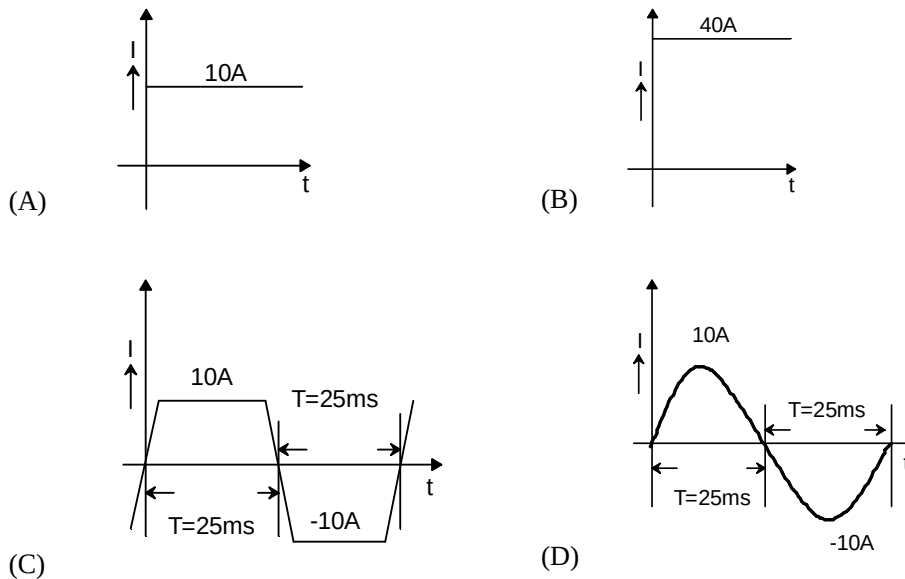


(b)

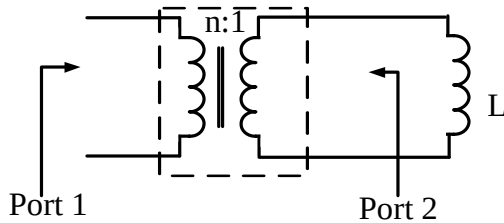
- Q.23 A steady dc current of 100 A is flowing through a power module (S, D) as shown in Figure (a). The V-I characteristics of the IGBT (S) and the diode (D) are shown in Figures (b) and (c), respectively. The conduction power loss in the power module (S, D), in watts, is _____.



- Q.24 A 4-pole, lap-connected, separately excited dc motor is drawing a steady current of 40 A while running at 600 rpm. A good approximation for the waveshape of the current in an armature conductor of the motor is given by



- Q.25 If an ideal transformer has an inductive load element at port 2 as shown in the figure below, the equivalent inductance at port 1 is



- (A) nL (B) n^2L (C) $\frac{n}{L}$ (D) $\frac{n^2}{L}$

Q. 26 – Q. 55 carry two marks each.

- Q.26 Candidates were asked to come to an interview with 3 pens each. Black, blue, green and red were the permitted pen colours that the candidate could bring. The probability that a candidate comes with all 3 pens having the same colour is _____.

- Q.27 Let $S = \sum_{n=0}^{\infty} n\alpha^n$ where $|\alpha| < 1$. The value of α in the range $0 < \alpha < 1$, such that $S = 2\alpha$ is _____.

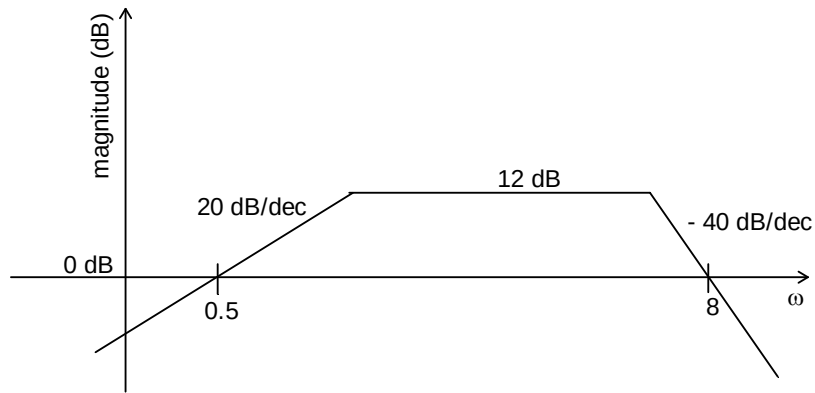
- Q.28 Let the eigenvalues of a 2×2 matrix A be 1, -2 with eigenvectors x_1 and x_2 respectively. Then the eigenvalues and eigenvectors of the matrix $A^2 - 3A + 4I$ would, respectively, be

- (A) 2, 14; x_1, x_2 (B) 2, 14; $x_1 + x_2, x_1 - x_2$
(C) 2, 0; x_1, x_2 (D) 2, 0; $x_1 + x_2, x_1 - x_2$

- Q.29 Let A be a 4×3 real matrix with rank 2. Which one of the following statement is TRUE?

- (A) Rank of $A^T A$ is less than 2.
(B) Rank of $A^T A$ is equal to 2.
(C) Rank of $A^T A$ is greater than 2.
(D) Rank of $A^T A$ can be any number between 1 and 3.

Q.30 Consider the following asymptotic Bode magnitude plot (ω is in rad/s).



Which one of the following transfer functions is best represented by the above Bode magnitude plot?

- (A) $\frac{2s}{(1+0.5s)(1+0.25s)^2}$
 (B) $\frac{4(1+0.5s)}{s(1+0.25s)}$
 (C) $\frac{2s}{(1+2s)(1+4s)}$
 (D) $\frac{4s}{(1+2s)(1+4s)^2}$

Q.31 Consider the following state-space representation of a linear time-invariant system.

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \mathbf{x}(t), \quad y(t) = \mathbf{c}^T \mathbf{x}(t), \quad \mathbf{c} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{x}(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

The value of $y(t)$ for $t = \log_e 2$ is _____.

Q.32

Loop transfer function of a feedback system is $G(s)H(s) = \frac{s+3}{s^2(s-3)}$. Take the Nyquist contour in the clockwise direction. Then, the Nyquist plot of $G(s)H(s)$ encircles $-1 + j0$

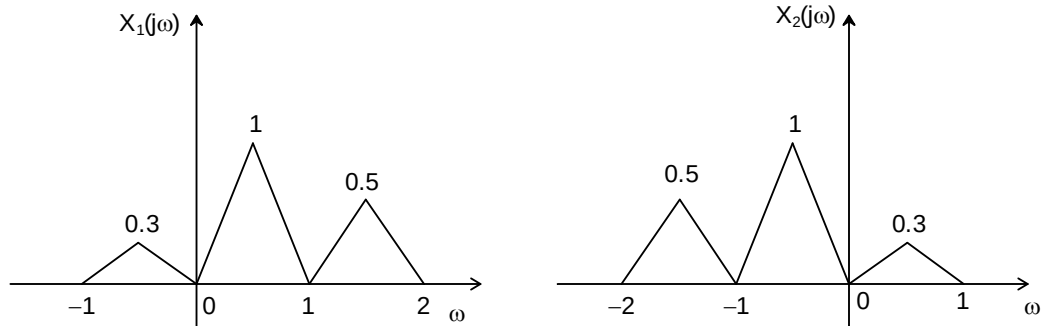
- (A) once in clockwise direction
 (B) twice in clockwise direction
 (C) once in anticlockwise direction
 (D) twice in anticlockwise direction

Q.33 Given the following polynomial equation

$$s^3 + 5.5s^2 + 8.5s + 3 = 0,$$

the number of roots of the polynomial, which have real parts strictly less than -1 , is _____.

Q.34 Suppose $x_1(t)$ and $x_2(t)$ have the Fourier transforms as shown below.



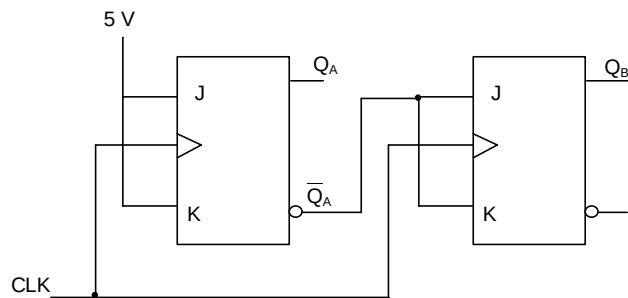
Which one of the following statements is TRUE?

- (A) $x_1(t)$ and $x_2(t)$ are complex and $x_1(t)x_2(t)$ is also complex with nonzero imaginary part
- (B) $x_1(t)$ and $x_2(t)$ are real and $x_1(t)x_2(t)$ is also real
- (C) $x_1(t)$ and $x_2(t)$ are complex but $x_1(t)x_2(t)$ is real
- (D) $x_1(t)$ and $x_2(t)$ are imaginary but $x_1(t)x_2(t)$ is real

Q.35 The output of a continuous-time, linear time-invariant system is denoted by $T\{x(t)\}$ where $x(t)$ is the input signal. A signal $z(t)$ is called eigen-signal of the system T , when $T\{z(t)\} = \gamma z(t)$, where γ is a complex number, in general, and is called an eigenvalue of T . Suppose the impulse response of the system T is real and even. Which of the following statements is TRUE?

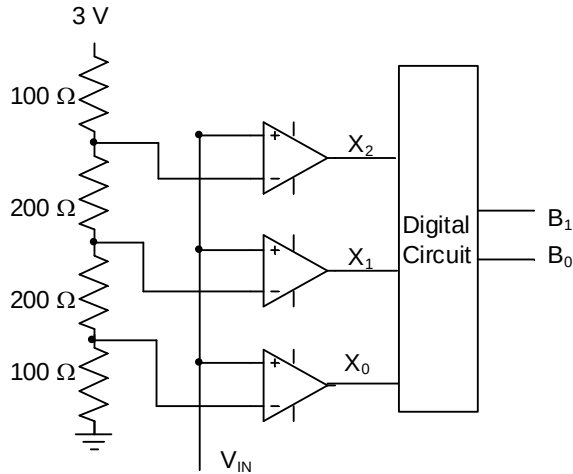
- (A) $\cos(t)$ is an eigen-signal but $\sin(t)$ is not
- (B) $\cos(t)$ and $\sin(t)$ are both eigen-signals but with different eigenvalues
- (C) $\sin(t)$ is an eigen-signal but $\cos(t)$ is not
- (D) $\cos(t)$ and $\sin(t)$ are both eigen-signals with identical eigenvalues

Q.36 The current state $Q_A Q_B$ of a two JK flip-flop system is 00. Assume that the clock rise-time is much smaller than the delay of the JK flip-flop. The next state of the system is

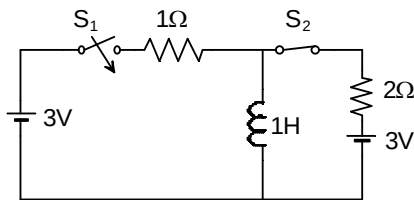


- (A) 00
- (B) 01
- (C) 11
- (D) 10

- Q.37 A 2-bit flash Analog to Digital Converter (ADC) is given below. The input is $0 \leq V_{IN} \leq 3$ Volts. The expression for the LSB of the output B_0 as a Boolean function of X_2 , X_1 , and X_0 is



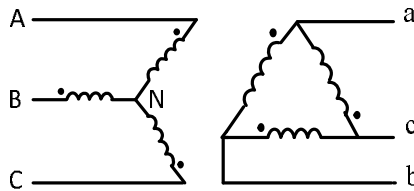
- (A) $X_0[\overline{X_2} \oplus \overline{X_1}]$ (B) $\overline{X_0}[\overline{X_2} \oplus X_1]$ (C) $X_0[X_2 \oplus X_1]$ (D) $\overline{X_0}[X_2 \oplus X_1]$
- Q.38 Two electric charges q and $-2q$ are placed at $(0,0)$ and $(6,0)$ on the x - y plane. The equation of the zero equipotential curve in the x - y plane is
- (A) $x = -2$ (B) $y = 2$ (C) $x^2 + y^2 = 2$ (D) $(x + 2)^2 + y^2 = 16$
- Q.39 In the circuit shown, switch S_2 has been closed for a long time. At time $t = 0$ switch S_1 is closed. At $t = 0^+$, the rate of change of current through the inductor, in amperes per second, is _____.



- Q.40 A three-phase cable is supplying 800 kW and 600 kVAr to an inductive load. It is intended to supply an additional resistive load of 100 kW through the same cable without increasing the heat dissipation in the cable, by providing a three-phase bank of capacitors connected in star across the load. Given the line voltage is 3.3 kV, 50 Hz, the capacitance per phase of the bank, expressed in microfarads, is _____.

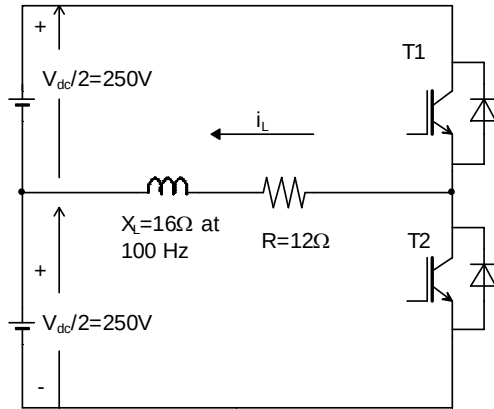
- Q.41 A 30 MVA, 3-phase, 50 Hz, 13.8 kV, star-connected synchronous generator has positive, negative and zero sequence reactances, 15%, 15% and 5% respectively. A reactance (X_n) is connected between the neutral of the generator and ground. A double line to ground fault takes place involving phases 'b' and 'c', with a fault impedance of $j0.1$ p.u. The value of X_n (in p.u.) that will limit the positive sequence generator current to 4270 A is _____.

- Q.42 If the star side of the star-delta transformer shown in the figure is excited by a negative sequence voltage, then

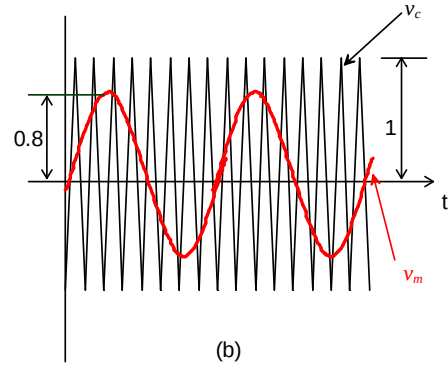


- (A) V_{AB} leads V_{ab} by 60°
 (B) V_{AB} lags V_{ab} by 60°
 (C) V_{AB} leads V_{ab} by 30°
 (D) V_{AB} lags V_{ab} by 30°
- Q.43 A single-phase thyristor-bridge rectifier is fed from a 230 V, 50 Hz, single-phase AC mains. If it is delivering a constant DC current of 10 A, at firing angle of 30° , then value of the power factor at AC mains is
- (A) 0.87 (B) 0.9 (C) 0.78 (D) 0.45

- Q.44 The switches T1 and T2 in Figure (a) are switched in a complementary fashion with sinusoidal pulse width modulation technique. The modulating voltage $v_m(t) = 0.8 \sin(200\pi t)$ V and the triangular carrier voltage (v_c) are as shown in Figure (b). The carrier frequency is 5 kHz. The peak value of the 100 Hz component of the load current (i_L), in ampere, is _____.

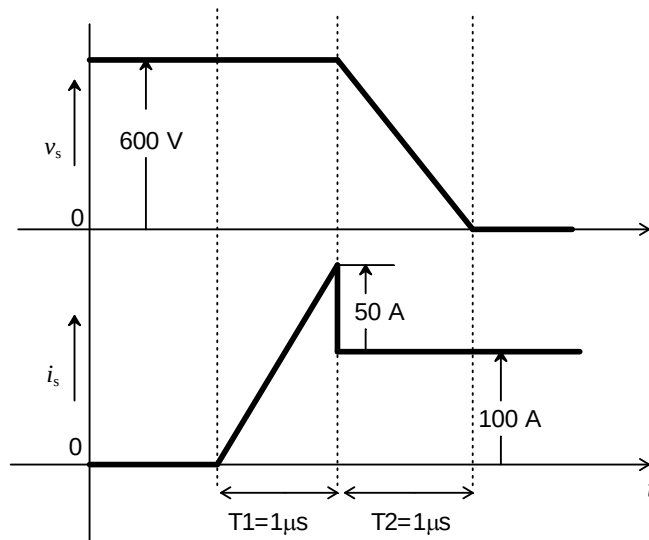


(a)



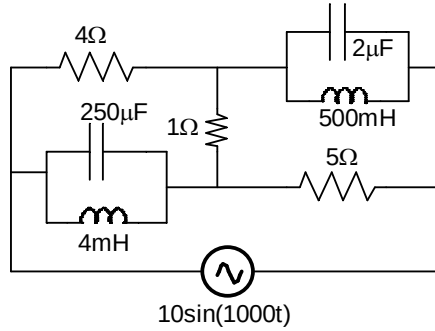
(b)

- Q.45 The voltage (v_s) across and the current (i_s) through a semiconductor switch during a turn-ON transition are shown in figure. The energy dissipated during the turn-ON transition, in mJ, is _____.



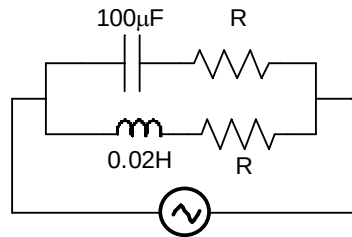
- Q.46 A single-phase 400 V, 50 Hz transformer has an iron loss of 5000 W at the rated condition. When operated at 200 V, 25 Hz, the iron loss is 2000 W. When operated at 416 V, 52 Hz, the value of the hysteresis loss divided by the eddy current loss is _____.
- Q.47 A DC shunt generator delivers 45 A at a terminal voltage of 220 V. The armature and the shunt field resistances are $0.01\ \Omega$ and $44\ \Omega$ respectively. The stray losses are 375 W. The percentage efficiency of the DC generator is _____.
- Q.48 A three-phase, 50 Hz salient-pole synchronous motor has a per-phase direct-axis reactance (X_d) of 0.8 pu and a per-phase quadrature-axis reactance (X_q) of 0.6 pu. Resistance of the machine is negligible. It is drawing full-load current at 0.8 pf (leading). When the terminal voltage is 1 pu, per-phase induced voltage, in pu, is _____.
- Q.49 A single-phase, 22 kVA, 2200 V/ 220 V, 50 Hz, distribution transformer is to be connected as an auto-transformer to get an output voltage of 2420 V. Its maximum kVA rating as an auto-transformer is
- (A) 22 (B) 24.2 (C) 242 (D) 2420
- Q.50 A single-phase full-bridge voltage source inverter (VSI) is fed from a 300 V battery. A pulse of 120° duration is used to trigger the appropriate devices in each half-cycle. The rms value of the fundamental component of the output voltage, in volts, is
- (A) 234 (B) 245 (C) 300 (D) 331
- Q.51 A single-phase transmission line has two conductors each of 10 mm radius. These are fixed at a center-to-center distance of 1 m in a horizontal plane. This is now converted to a three-phase transmission line by introducing a third conductor of the same radius. This conductor is fixed at an equal distance D from the two single-phase conductors. The three-phase line is fully transposed. The positive sequence inductance per phase of the three-phase system is to be 5% more than that of the inductance per conductor of the single-phase system. The distance D, in meters, is _____.

- Q.52 In the circuit shown below, the supply voltage is $10 \sin(1000t)$ volts. The peak value of the steady state current through the 1Ω resistor, in amperes, is _____.

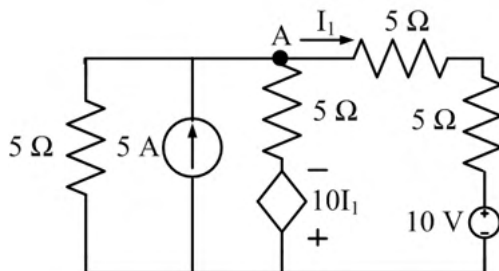


- Q.53 A dc voltage with ripple is given by $v(t) = [100 + 10 \sin(\omega t) - 5 \sin(3\omega t)]$ volts. Measurements of this voltage $v(t)$, made by moving-coil and moving-iron voltmeters, show readings of V_1 and V_2 respectively. The value of $V_2 - V_1$, in volts, is _____.

- Q.54 The circuit below is excited by a sinusoidal source. The value of R , in Ω , for which the admittance of the circuit becomes a pure conductance at all frequencies is _____.



- Q.55 In the circuit shown below, the node voltage V_A is _____ V.



END OF THE QUESTION PAPER

Q. No	Type	Section	Key	Marks
1	MCQ	GA	B	1
2	MCQ	GA	A	1
3	MCQ	GA	C	1
4	MCQ	GA	D	1
5	MCQ	GA	B	1
6	MCQ	GA	A	2
7	MCQ	GA	D	2
8	MCQ	GA	D	2
9	MCQ	GA	C	2
10	MCQ	GA	C	2
1	NAT	EE-1	0.0 : 0.0	1
2	NAT	EE-1	3.0 : 3.0	1
3	MCQ	EE-1	A	1
4	MCQ	EE-1	B	1
5	MCQ	EE-1	B	1
6	MCQ	EE-1	B	1
7	MCQ	EE-1	A	1
8	MCQ	EE-1	C	1
9	MCQ	EE-1	A	1
10	MCQ	EE-1	B	1
11	MCQ	EE-1	D	1
12	NAT	EE-1	18.0 : 20.0	1
13	MCQ	EE-1	B	1
14	NAT	EE-1	99.0 : 101.0	1
15	MCQ	EE-1	D	1
16	MCQ	EE-1	C	1
17	NAT	EE-1	1.9 : 2.1	1
18	NAT	EE-1	0.5 : 0.5	1
19	MCQ	EE-1	B	1
20	NAT	EE-1	0.83 : 0.85	1
21	MCQ	EE-1	A	1
22	NAT	EE-1	0.39 : 0.41	1
23	NAT	EE-1	169.0 : 171.0	1
24	MCQ	EE-1	C	1
25	MCQ	EE-1	B	1
26	NAT	EE-1	0.2 : 0.2	2
27	NAT	EE-1	0.28 : 0.31	2
28	MCQ	EE-1	A	2
29	MCQ	EE-1	B	2
30	MCQ	EE-1	A	2
31	NAT	EE-1	5.9 : 6.1	2
32	MCQ	EE-1	A	2
33	NAT	EE-1	2.0 : 2.0	2
34	MCQ	EE-1	C	2
35	MCQ	EE-1	D	2
36	MCQ	EE-1	C	2
37	MCQ	EE-1	A	2
38	MCQ	EE-1	D	2
39	NAT	EE-1	1.9 : 2.1	2

40	NAT	EE-1	47.0 : 49.0	2
41	NAT	EE-1	1.05 : 1.15	2
42	MCQ	EE-1	D	2
43	MCQ	EE-1	C	2
44	NAT	EE-1	9.9 : 10.1	2
45	NAT	EE-1	74.0 : 76.0	2
46	NAT	EE-1	1.4 : 1.5	2
47	NAT	EE-1	86.0 : 88.0	2
48	NAT	EE-1	1.58 : 1.62	2
49	MCQ	EE-1	C	2
50	MCQ	EE-1	A	2
51	NAT	EE-1	1.41 : 1.47	2
52	NAT	EE-1	1.0 : 1.0	2
53	NAT	EE-1	0.30 : 0.33	2
54	NAT	EE-1	14.0 : 14.2	2
55	NAT	EE-1	11.25 : 11.50	2